

LF Energy TAC p-SWAMP review 8th September 2026





OLF ENERGY

p-SWAMP Partners



SINTEF



NTNU



RI.
SE

lede

OLF ENERGY

Statnett



The Research
Council of Norway

Reference Groupe



GE VERNOVA

FINGRID

LANDSNET

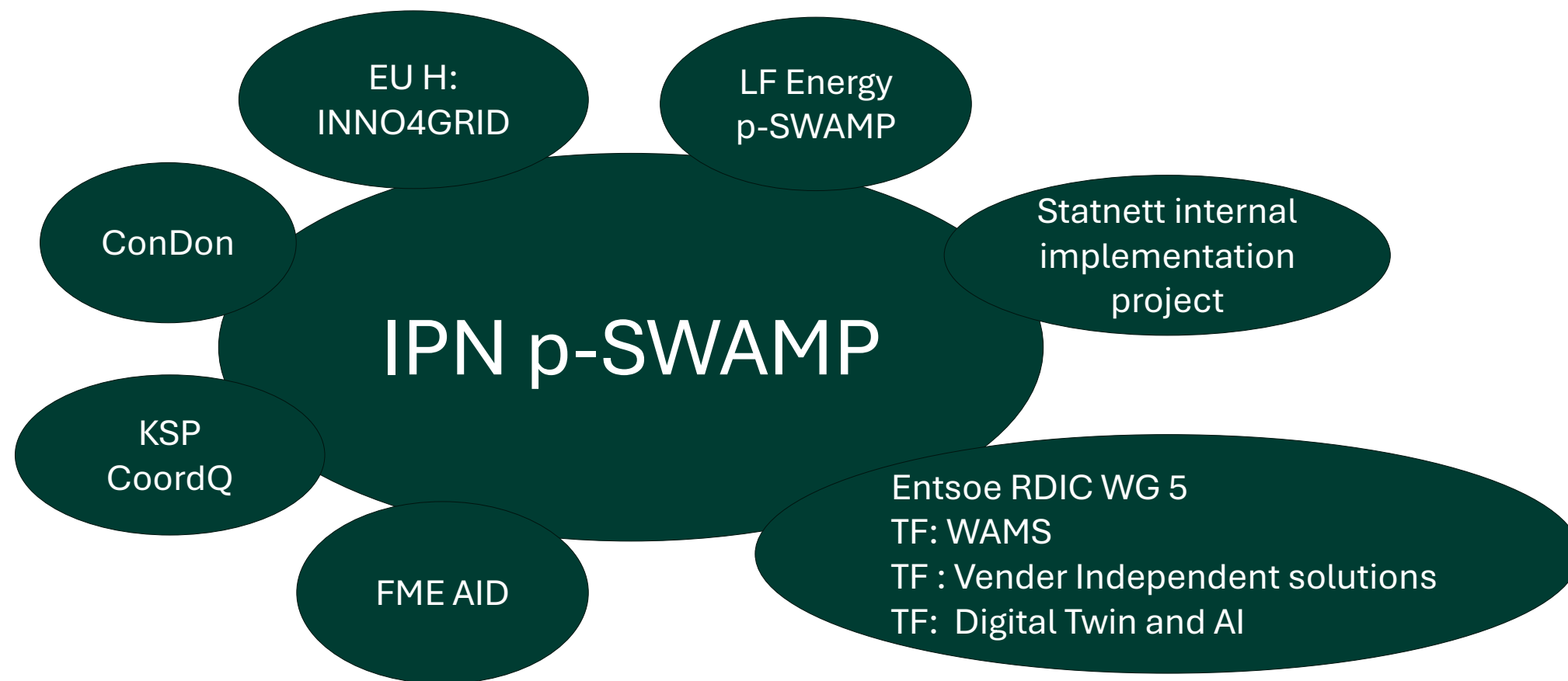


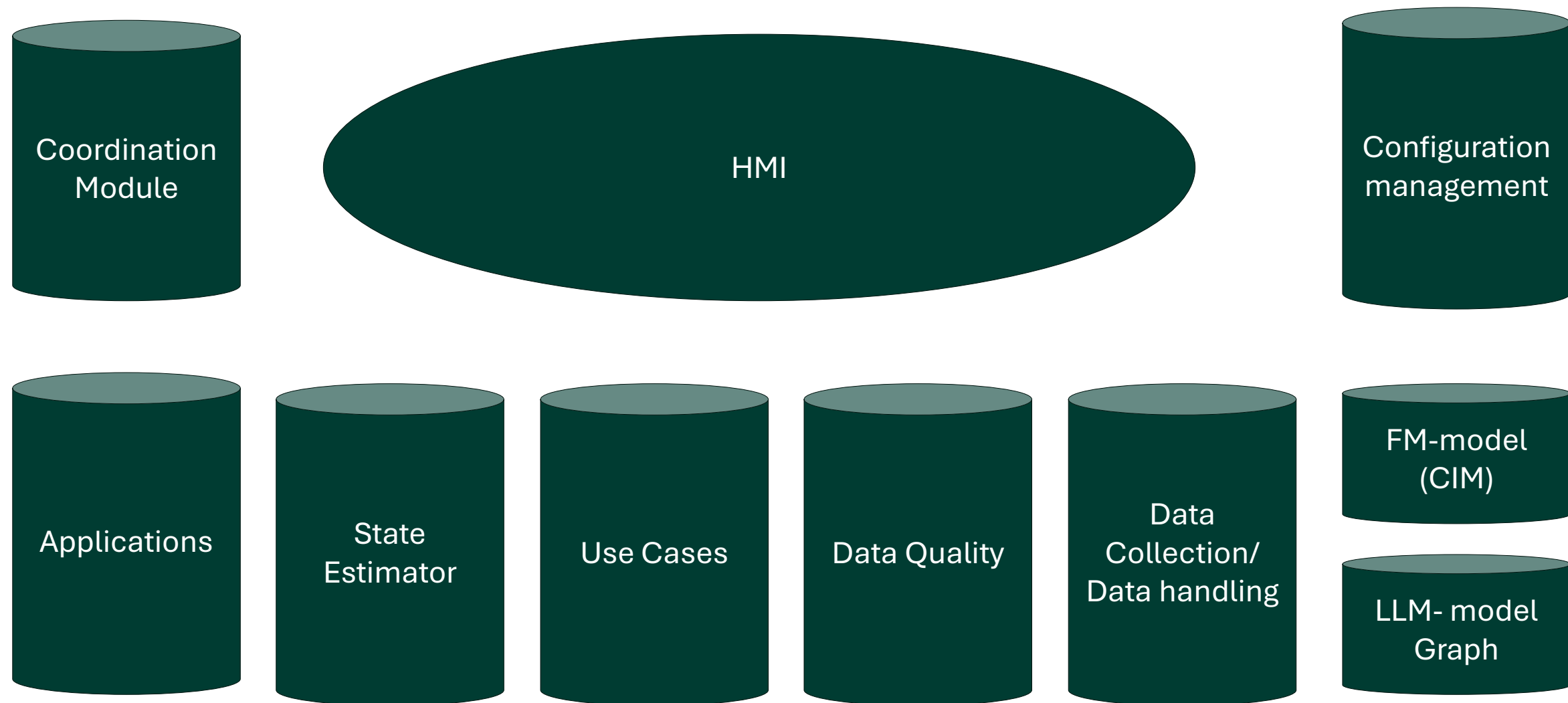
Statkraft



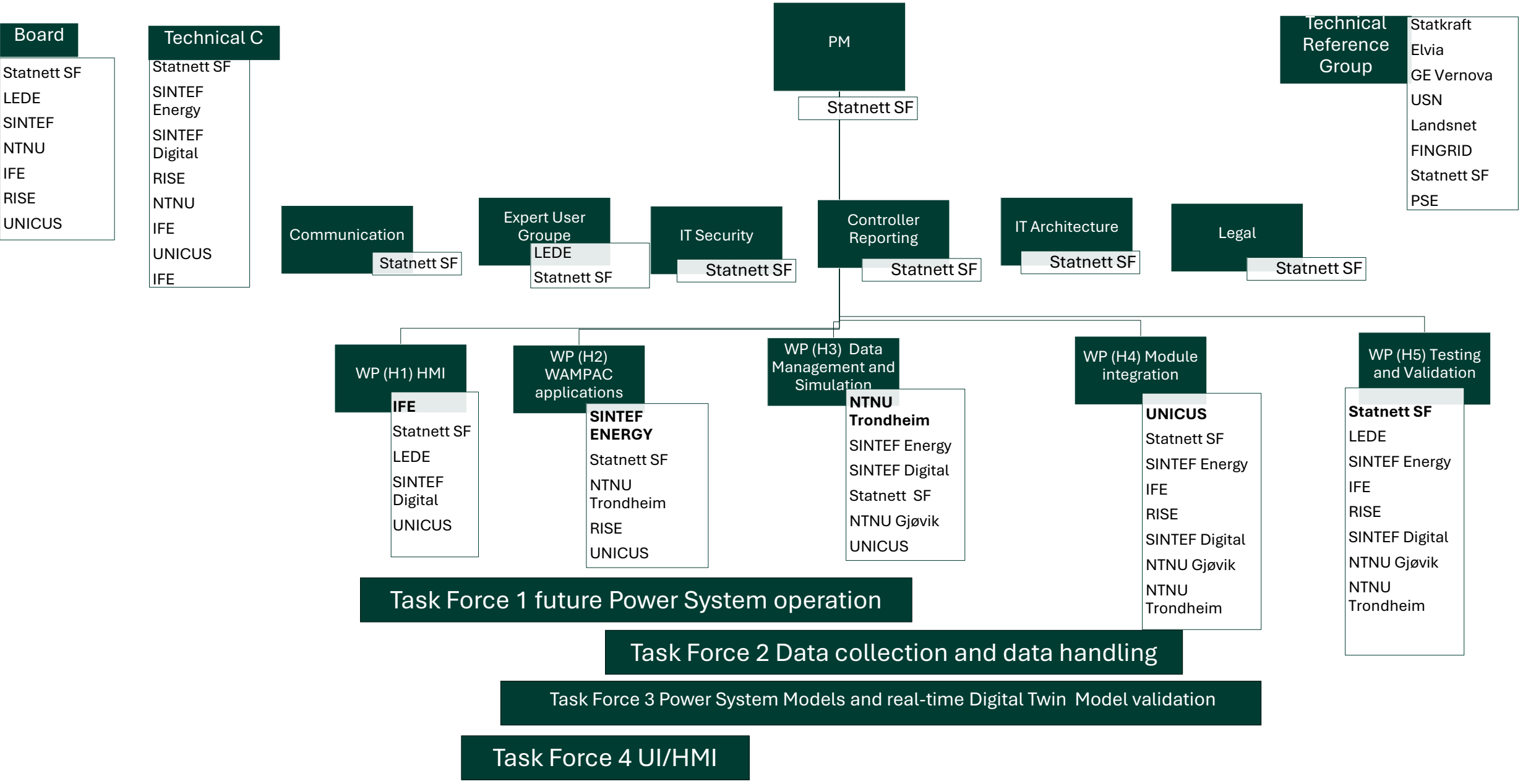
Glvia

Interfaces





Project Organization



H0:
Management
communication,
dissemination
(Statnett)

H1: Advanced Web based HMI (IFE)

T1.1 HMI specification
definition
(SINTEF D, IFE, LEDE,
Statnett)

T1.2 Conceptual HMI
development
(IFE, SINTEF D, LEDE,
Statnett)+ NTNU+ SINTEF E.

T1.3 Web based
front-end
implementation
(IFE, UNICUS) + lede
+statnett

T1.4 FM-based agent
implementation
(SINTEF D, UNICUS) +
lede +statnett

H2: WAMPAC applications for operation support (SINTEF E)

T2.2 Coordinated
voltage control
(NTNU T, SINTEF E,
Statnett)+UNICUS

T2.3 Contingency
management
SINTEF E, NTNU T,
Statnett+UNICUS

T2.4 Real time
recommendation of
preventive and mitigation
actions (SINTEF E, NTNU T,
Statnett)+UNICUS

WAMPAC algorithms implementation

- Inter-area oscillation detection,
- Voltage instability classification
- SPS coordination

(NTNU T,SINTEF E, RISE)

T2.1 Microservice-based architecture (SINTEF E, Statnett, NTNU T, Lede, UNICUS)

H3: Data conditioning and linear state estimation (NTNU T)

T3.1 Data collection and
harmonization (Statnett, UNICUS,
SINTEF E, NTNU T, NTNU G)

T3.3 Implementation AI based
anomaly detection (NTNU G,
SINTEF D, SINTEF E)

T3.4 Linear state estimation
(NTNU T, Statnett, SINTEF E)

T3.2 Extension for the Statnett Information Model (SINTEF D, Statnett)

H4: Modules integration and deployment (UNICUS)

T4.1 Integration HMI modules
(UNICUS, IFE, SINTEF D)

T4.2 Integration WAMPAC modules
(UNICUS, RISE,SINTEF E, NTNU T)

T4.3 Integration data conditioning
modules (UNICUS, NTNU T, NTNU G,
SINTEF D, SINTEF E)

H5:
Module
validations and
pilot
demonstration
(Statnett)

T5.1 Pilot
configuration
(UNICUS, Statnett,
Lede)

T5.2 Module
validation with
online data
(SINTEF E, ALL)

T5.3 Pilot
demonstration
(Statnett, Lede,
UNICUS)

T5.4 Roadmap for
technology
transfer (Statnett,
Lede)

Develop User
stories (Statnett,
Lede)

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